Water Quality Report

Demo Public Dock Buoy · 2026-10-03 to 2026-10-10

Normal

Coordinates	33.6048° N, 117.8835° W (Seaview Dock CA)
Water Body Type	Marine (from device registry)
Client / Contract	Not available from device registry
Report Date	2026-10-10
Last Reading	2026-10-10 08:41 UTC (3 minutes before this report)
Prepared By	Clean Earth Rovers

1. Summary

- Overall status: Normal — no action required at this time.
- Every parameter with a site baseline held within it for the 2026-10-03 to 2026-10-10 reporting period.
- No pollution event signatures were identified; diel and tidal rhythms tracked the site baseline throughout.
- Recommendation: continue routine monitoring.

2. Parameter Data

Parameter	Site Baseline	Min	Max	Mean	Median	Out of range	Flag
Dissolved Oxygen (mg/L)	Not established (site-specific)	0.00	11.15	8.40	8.34	—	Not assessed
ORP (mV)	-200-400	180.20	299.70	240.11	238.40	0%	Normal
pH	Not established (site-specific)	7.82	8.39	8.10	8.10	—	Not assessed
Conductivity (µS/cm)	40000-55000	44038	49994	47052	46843	0%	Normal
Temperature (°F)	50-80	64.93	78.76	71.71	71.44	0%	Normal

Flag values: Normal, Elevated, Low, or Exceedance relative to the site baseline. Out of range is the share of the period's series buckets whose average sat outside that baseline — approximate, at bucket resolution, and shown because a flag alone cannot separate a single stray reading from a sustained offset.
Site Baseline for ORP (mV), Conductivity (µS/cm), Temperature (°F) is this device's operator-configured threshold from the device registry — an alert limit the operator set, not a scientifically reviewed range. A parameter with no usable registry threshold shows no baseline at all ("Not established") rather than falling back to a fixed table.

3. Parameter Analysis

Each chart plots the period's series bucket means, oldest to newest. The shaded band is the site baseline where one exists; the dots mark the plotted high and low.

Dissolved Oxygen (mg/L)

Not assessed

Followed a clear diel rhythm. No baseline is established for this parameter -- reported for reference only, not flagged against a range. Not assessed: this device's dissolved oxygen thresholds fall outside a plausible range (0 to 30), so they read as a placeholder rather than a configured limit.

pH

Not assessed

Followed a clear diel rhythm. No baseline is established for this parameter -- reported for reference only, not flagged against a range. Not assessed: this device's pH thresholds fall outside a plausible range (0 to 14), so they read as a placeholder rather than a configured limit.

Temperature (°F)

Normal

Followed a clear diel rhythm, remaining within the 50-80 °F operator-set threshold for this device. Operator-set threshold for this device (50-80 °F, from the device registry).

4. Data Quality

Check	Result	Notes
Data completeness	100.0%	169 of 169 readings returned for the period were usable for at least one parameter. 0 readings were left out because no requested parameter was usable. Values from likely failed sensors were left out. Values outside plausibility bounds may be sensor faults and were left out. no usable readings for: Turbidity (Relative). Share of returned readings, not coverage against an expected sampling cadence.
Calibration status	Review	Turbidity has a sustained zero-variance run at exactly 0 or 1005, indicating a likely failed sensor. Affected readings were excluded; inspect the sensor. Readings outside plausibility bounds without a fault flag (pH: 1; Conductivity (µS/cm): 4) may be sensor faults and were left out. Inspect and recalibrate the affected probes.
Drift indicators	Not assessed	Not assessed — this pipeline has no drift detector. Requires comparison against a calibration record, which is not available from the device API.
Biofouling indicators	Not assessed	Not assessed — this pipeline has no biofouling detector. Requires service history or a fouling-sensitive baseline, neither of which is available here.
Sensor agreement	Not assessed	Not assessed — cross-sensor agreement needs a second co-located pod or a grab-sample result to compare against.

5. Recommendations

Operational: No action needed; maintain routine calibration schedule.

Investigative: None required this period.

Stakeholder: Routine report distribution to client only.